EC442 Macroeconomics for MRes students
London School of Economics

Instructor: Matthias Doepke
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Problem Set 3 Solutions

Enrico D. Turri[1]

Dynamic Programming: Practice

Question 1: In this question, you will use polynomial approximation to numerically solve the planning problem from Question 1 of Problem Set 2. Consider the Bellman equation for the social planning problem for the "standard model:"

$$v(k) = \max_{k'} \left\{ \log\left((1-\delta)k + Ak^{\alpha} - k'\right) + \beta v(k') \right\}.$$

We will once again use the parameter values $\beta = 0.9$, $A = 3$, $\alpha = 0.3$, and $\delta = 0.2$.

- We will approximate the value function by polynomials of the form:

$$v(k) = \sum_{i=0}^{n} a_i k^i.$$

  The objective is to find coefficients a_i that yield a good approximation to the value function defined by the Bellman equation for a given order n of the polynomial.

- Start by fixing n and making an initial guess for the coefficients $a_{i,0} = 0$ for all i, so that the initial guess for the value function is:

$$v_0(k) = \sum_{i=0}^{n} a_{i,0} k^i = 0.$$

- Then use the Bellman equation (just as in discrete-state dynamic programming) to compute a new guess $v_1(k)$ for the value function, i.e., a new set of coefficients $a_{i,1}$. What is convenient about using a parametric function to approximate the value function is that we need to evaluate the Bellman equation only at a few points to make a new guess. Specifically, we need to know $v_1(k)$ at $n+1$ distinct points (which we will refer to as "nodes") to be able to solve for the coefficients $a_{i,1}$. We therefore first need to fix a set of nodes K. As a first pass, use at least $n+1$ evenly spaced points in the interval $[1, 10]$ as your set of nodes (this choice

[1] e.turri@lse.ac.uk

can in principle be optimized). For each capital value $k \in K$ at a node, compute a new guess for the value function by solving:

$$v_1(k) = \max_{k'} \left\{ \log((1-\delta)k + Ak^\alpha - k') + \beta v_0(k') \right\}.$$

It is up to you to decide how to carry out the maximization step. You could do something simple such as a grid search, or use an existing routine such as `fminbnd` or `fmincon` in MATLAB or something similar in your language of choice.

- The maximization step results in one value for the new guess of the value function $v_1(k)$ at each of the nodes $k \in K$. Use these values to find new polynomial coefficients $a_{i,1}$ that fit these points. Since the coefficients enter linearly, with exactly $n+1$ nodes the polynomial interpolates the points and the coefficients solve a system of $n+1$ linear equations; with more nodes, they solve a linear least-squares problem. The backslash operator in MATLAB handles both cases, as does `numpy.linalg.lstsq` in Python.

- Continue constructing new guesses of the value function in this way, until the convergence criterion:
$$\max_i |a_{i,j+1} - a_{i,j}| < 10^{-5}$$
is met, where j indexes the iterations.

Compute polynomial approximations to the value function for $n = 2, 3, \ldots, 6$:

1. Plot the resulting approximations on the interval $k \in [0, 15]$ together with the computed value function using the Howard algorithm from Problem Set 2.

2. Provide a second plot of the approximated policy functions, once again together with the policy function resulting from the Howard improvement algorithm.

3. How large does the order of the polynomial have to be to yield a reasonable approximation?

Answer: Figures 1 and 2 show the approximated value and policy functions for $n = 2, 3, \ldots, 6$, together with those computed with Howard's algorithm on a grid. The shaded area is the interval $[1, 10]$ spanned by the nodes. Clearly, the higher the degree of the polynomial, the closer we can fit the value function. We see that the value and policy functions overlap almost perfectly when using the fourth order polynomial. This is of course only true for the part of the grid that we use to estimate it, the interval $[1, 10]$ spanned by the nodes. Outside this interval, the polynomial extrapolates, and the approximation quickly deteriorates. Note that extrapolation is not confined to the

plots: it already happens inside the iteration. At $k = 10$, resources are $(1-\delta)k + Ak^{\alpha} = 0.8 \cdot 10 + 3 \cdot 10^{0.3} \approx 14$, so the maximization step evaluates $v_j(k')$ at values of k' well above the largest node.

Additional remarks

Two questions to think about. (i) What is particularly worrying about the policy function for $n = 3$ in Figure 2? (ii) Does the same problem arise whenever we use a polynomial of odd degree?

Answers: (i) With a third-degree polynomial, the approximated value function becomes convex for k above about 8.5, because the leading coefficient a_3 is positive. A convex continuation value rewards saving more and more at the margin, so the policy function jumps: for k above about 12, it lies above the 45-degree line, and the planner chooses the largest capital stock allowed, $k' = 15$. The approximation therefore creates a spurious second steady state at about 12, which is unstable: an economy that starts above it accumulates capital until it hits the bound, instead of converging to the steady state $k_{ss} = 4.56$. What makes this particularly worrying is that nothing warns us. The coefficients converge, and the value function looks reasonable, but the policy function, which is what determines the dynamics, is qualitatively wrong.

(ii) Yes. With exactly $n + 1$ nodes, the leading coefficient a_n of the interpolating polynomial is the n-th divided difference of the data, which equals $v^{(n)}(\xi)/n!$ for some $\xi \in [1, 10]$. The value function behaves like a logarithm, whose derivatives alternate in sign: the n-th derivative of $\log(k)$ is $(-1)^{n-1}(n-1)!/k^n$. Hence $a_n > 0$ for odd n and $a_n < 0$ for even n. With $a_n > 0$, the polynomial eventually becomes increasing and convex, and the policy function jumps as in the panels for $n = 3$ and $n = 5$ of Figure 2; numerically, this happens for $n = 3$, 5, and 7. With $a_n < 0$, the polynomial eventually becomes decreasing and concave, so the policy function bends below the true one at high k, as in the panels for $n = 2$, 4, and 6 of Figure 2. This is also wrong, but less harmful, since the policy function stays below the 45-degree line. The root of the problem is not the parity of the degree, though: no polynomial of degree two or higher is increasing and concave on all of $[0, \infty)$, because its derivative would then be a nonincreasing polynomial bounded below by zero, hence constant. Extrapolation therefore always fails, and the sign of a_n only decides how. A simple remedy is to never extrapolate: restrict k' to the interval spanned by the nodes, $[1, 10]$, which does not bind at the optimum, since the true policy function maps $[1, 10]$ into itself. Alternatively, one can approximate the value function by a polynomial in $\log(k)$, which captures its shape much better: with a third-degree polynomial in $\log(k)$, the largest error in the policy function on $[1, 10]$ falls from about 0.55 to about 0.01.

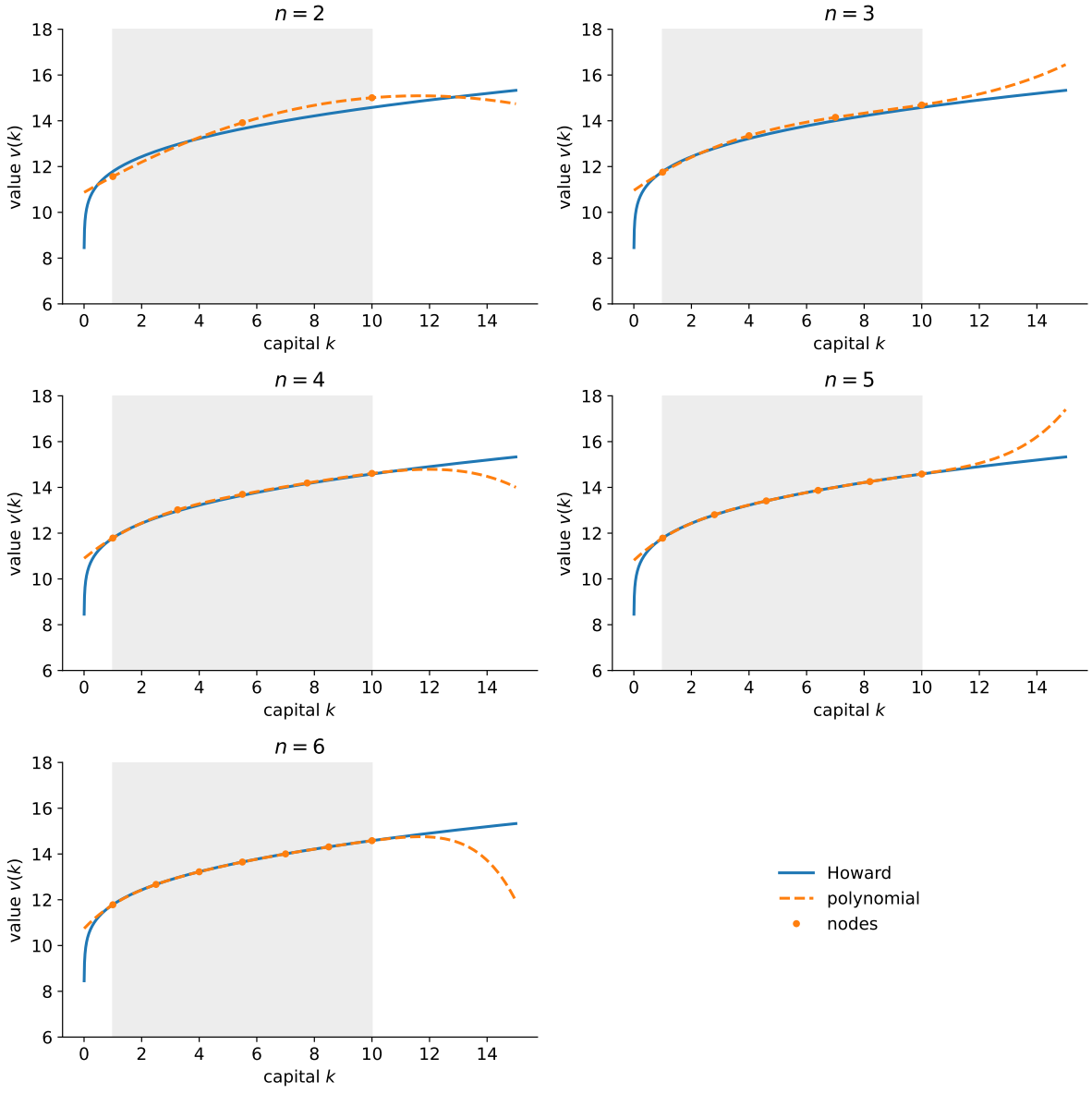


Figure 1: Value function: polynomial approximation of order n and Howard's algorithm. The shaded area is the interval spanned by the nodes.

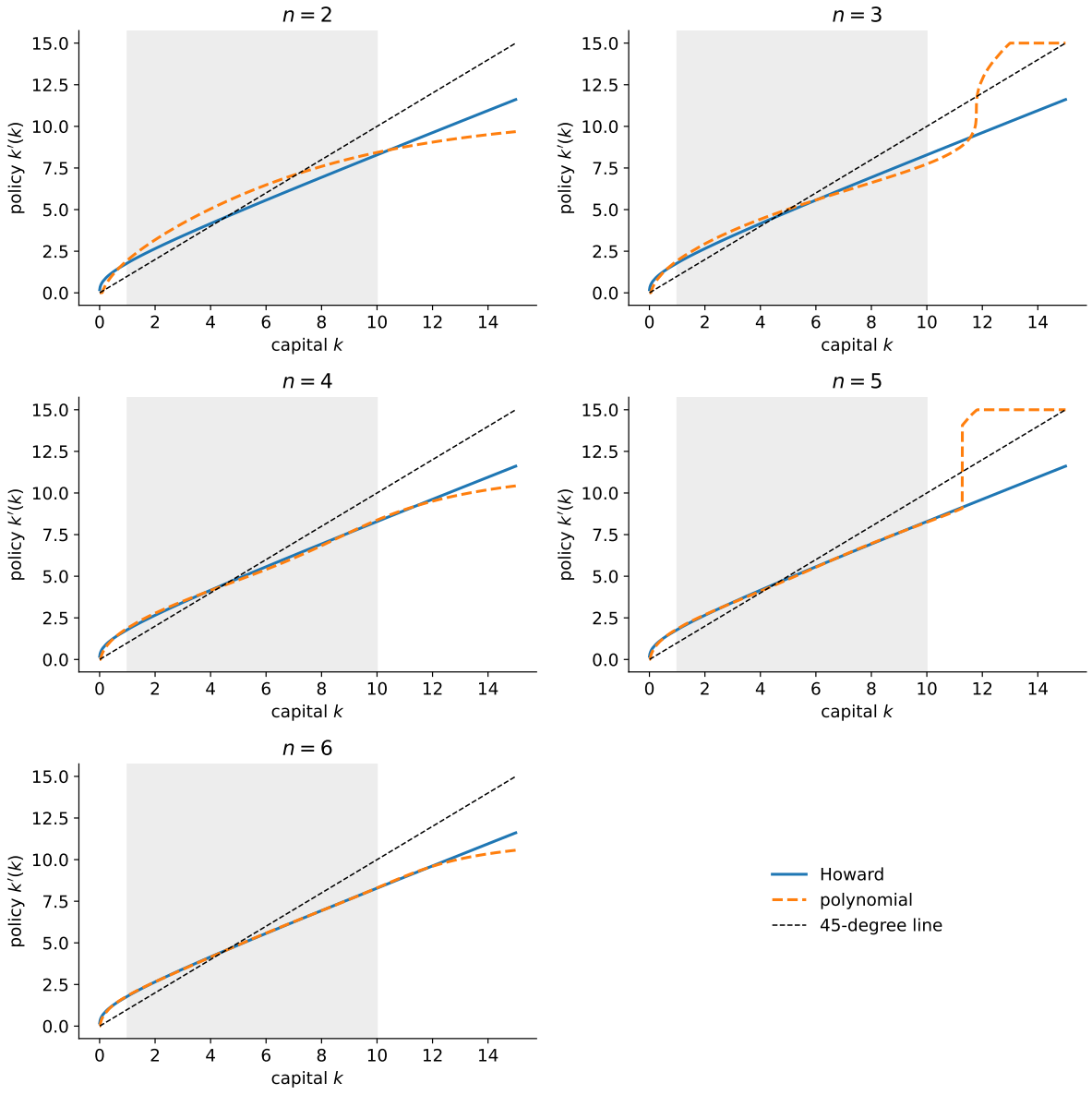


Figure 2: Policy function: polynomial approximation of order n and Howard's algorithm. The shaded area is the interval spanned by the nodes.

A Model with Human Capital

Question 2: Consider an economy in which the representative consumer has preferences:

$$\sum_{t=0}^{\infty} \beta^t \frac{c_t^{1-\sigma}}{1-\sigma},$$

where $\sigma > 0$ and $\sigma \neq 1$.

The production function is given by:

$$y_t = A k_t^{\alpha} h_t^{1-\alpha}.$$

The laws of motion for physical and human capital are:

$$k_{t+1} = (1-\delta)k_t + x_{kt},$$
$$h_{t+1} = (1-\delta)h_t + x_{ht}.$$

Output can be converted one-to-one into consumption, physical investment, or human-capital investment, so that the resource constraint is

$$c_t + x_{kt} + x_{ht} = y_t.$$

Assume that $\beta \left(A\alpha^{\alpha}(1-\alpha)^{1-\alpha} + 1 - \delta \right)^{1-\sigma} < 1$.

1. Formulate the Bellman equation for the social planning problem, and show that the value function is homogeneous (of which degree?).

   Answer: The resource constraint can be written as

   $$c_t + k_{t+1} + h_{t+1} = (1-\delta)(k_t + h_t) + A k_t^{\alpha} h_t^{1-\alpha}.$$

   From there, we can write the Bellman equation as

   $$v(k,h) = \max_{k',h'} \left\{ \frac{\left(A k^{\alpha} h^{1-\alpha} + (1-\delta)(k+h) - k' - h' \right)^{1-\sigma}}{1-\sigma} + \beta v(k',h') \right\}.$$

   Also, by definition,

   $$v(k,h) = \max_{\{k_{t+1}, h_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t \frac{\left(A k_t^{\alpha} h_t^{1-\alpha} + (1-\delta)(k_t + h_t) - k_{t+1} - h_{t+1} \right)^{1-\sigma}}{1-\sigma}$$
   $$\text{s.t.} \qquad k_0 = k, \quad h_0 = h.$$

   We need to show that, for any $\lambda > 0$, $v(\lambda k, \lambda h) = \lambda^r v(k,h)$ for some r. Take any sequence $\{k_{t+1}, h_{t+1}\}_{t=0}^{\infty}$ that is feasible from $(k_0, h_0) = (k,h)$, with consumption

c_t, and scale it by λ. Since the production function has constant returns to scale and the rest of the resource constraint is linear, consumption scales by λ too:

$$A(\lambda k_t)^{\alpha}(\lambda h_t)^{1-\alpha} + (1-\delta)(\lambda k_t + \lambda h_t) - \lambda k_{t+1} - \lambda h_{t+1} = \lambda c_t.$$

Hence the scaled sequence is feasible from $(\lambda k, \lambda h)$. Since $(\lambda c_t)^{1-\sigma} = \lambda^{1-\sigma} c_t^{1-\sigma}$, its utility is $\lambda^{1-\sigma} U$, where U is the utility of the original sequence. The value $v(\lambda k, \lambda h)$ is the supremum of utility over all sequences feasible from $(\lambda k, \lambda h)$, so it is at least $\lambda^{1-\sigma} U$. This holds for every sequence feasible from (k, h), so taking the supremum over them, and using $\lambda^{1-\sigma} > 0$,

$$v(\lambda k, \lambda h) \geq \lambda^{1-\sigma} v(k, h) \qquad \text{for all } \lambda > 0 \text{ and all } (k, h).$$

Applying this inequality at the point $(\lambda k, \lambda h)$, with the factor $1/\lambda$, gives

$$v(k, h) \geq \lambda^{\sigma-1} v(\lambda k, \lambda h), \qquad \text{that is,} \qquad v(\lambda k, \lambda h) \leq \lambda^{1-\sigma} v(k, h).$$

Combining the two inequalities,

$$v(\lambda k, \lambda h) = \lambda^{1-\sigma} v(k, h).$$

Here we use the condition in the question, which ensures that $v(k, h)$ is finite. Hence, the value function $v(k, h)$ is homogeneous of degree $1 - \sigma$.

2. Compute the balanced growth rate of the economy, and characterize the transitional dynamics for arbitrary initial levels of k_0 and h_0.

   Answer: Taking the first order conditions of the Bellman equation yields

$$c^{-\sigma} = \beta v_k(k', h')$$
$$c^{-\sigma} = \beta v_h(k', h')$$

such that $v_k(k', h') = v_h(k', h')$. The envelope condition yields

$$v_k(k, h) = c^{-\sigma} \left(\alpha A k^{\alpha-1} h^{1-\alpha} + 1 - \delta \right)$$
$$v_h(k, h) = c^{-\sigma} \left((1-\alpha) A k^{\alpha} h^{-\alpha} + 1 - \delta \right)$$

such that

$$\frac{\alpha}{1-\alpha} h' = k'.$$

That is, the ratio between physical and human capital is constant. Combining the first order condition, the envelope condition, and the optimal human/physical capital ratio, we get the growth rate of consumption:

$$\left(\frac{c'}{c} \right)^{\sigma} = \beta \left(A \alpha^{\alpha} (1-\alpha)^{1-\alpha} + 1 - \delta \right).$$

To investigate transitional dynamics, note that for any k_0 and h_0, the ratio $k_t/h_t = \alpha/(1-\alpha)$ holds for all $t \geq 1$. This relies on investment being reversible: nothing restricts the sign of x_{kt} and x_{ht}, so the planner can move from any (k_0, h_0) to the optimal ratio in a single period, if necessary by disinvesting one type of capital. Let $K = k + h$ denote total capital. At the optimal ratio, $k = \alpha K$ and $h = (1-\alpha)K$, so that output is $\tilde{A}K$, with $\tilde{A} \equiv A\alpha^{\alpha}(1-\alpha)^{1-\alpha}$, and the resource constraint becomes $c + K' = (\tilde{A} + 1 - \delta)K$. This means that for $t = 1$ onward, the Bellman equation reduces to

$$V(K) = \max_{K'} \left\{ \frac{\left((\tilde{A} + 1 - \delta)K - K'\right)^{1-\sigma}}{1-\sigma} + \beta V(K') \right\}.$$

This problem is the same as the AK model that we solved in problem set 1. Remember that we found there that optimal saving and consumption are a constant fraction of output. Hence, from period 1 onward, consumption, output, and physical and human capital must all grow at the same rate given by the Euler equation above. That is,

$$\frac{c_{t+1}}{c_t} = \frac{k_{t+1}}{k_t} = \frac{h_{t+1}}{h_t} = \frac{y_{t+1}}{y_t} = \left(\beta\left(\tilde{A} + 1 - \delta\right)\right)^{1/\sigma} \quad \text{for all } t \geq 1.$$

To characterize the transition, note that from period 1 onward, the optimal policy will be that $c = (1-s)(\tilde{A} + 1 - \delta)K$, where $s = \beta^{1/\sigma}(\tilde{A} + 1 - \delta)^{(1-\sigma)/\sigma}$ is the saving rate derived in problem set 1. Note also that the Euler equation needs to hold from period 0 onward. Hence,

$$\frac{c_1}{c_0} = \left(\beta\left(\tilde{A} + 1 - \delta\right)\right)^{1/\sigma} = \frac{(1-s)(\tilde{A} + 1 - \delta)K_1}{Ak_0^{\alpha}h_0^{1-\alpha} + (1-\delta)(k_0 + h_0) - K_1},$$

from where we can solve for K_1, and hence $k_1 = \alpha K_1$ and $h_1 = (1-\alpha)K_1$. Intuitively, if $k_0 > \alpha h_0/(1-\alpha)$, then in period 0 investment in human capital must be large relative to investment in physical capital.

Additional remarks

A question to think about. Is the value function still homogeneous with logarithmic utility, $\sigma = 1$?

Answer: No: scaling consumption by λ adds $\log(\lambda)$ to utility in every period, so that $v(\lambda k, \lambda h) = v(k, h) + \log(\lambda)/(1-\beta)$.

An AK Model with Endogenous Labor Supply

Question 3: Consider an economy in which the representative consumer has preferences:

$$\sum_{t=0}^{\infty} \beta^t (c_t + \log(1 - n_t)),$$

where c_t is consumption, n_t is labor supply, and $1 - n_t$ is leisure. Capital k_t depreciates completely every period. There are separate technologies for producing capital and the consumption good; in particular, capital is produced using capital and labor, whereas consumption is produced using only capital:

$$c_t = Ak_t,$$
$$k_{t+1} = k_t n_t.$$

Notice that the same capital is used in both sectors; the consumer only has to decide on labor supply, but not on the allocation of capital. The consumer also has to obey the constraint $0 \leq n_t < 1$, and the initial capital stock $k_0 > 0$ is given.

1. Formulate the Bellman equation for the social planning problem.

   Answer: We can write the Bellman equation as

   $$v(k) = \sup_{0 \leq n < 1} \{Ak + \log(1 - n) + \beta v(kn)\}$$

   or, equivalently, for $k > 0$,

   $$v(k) = \sup_{0 \leq k' < k} \left\{ Ak + \log\left(1 - \frac{k'}{k}\right) + \beta v(k') \right\}.$$

   We write sup rather than max because the choice set $[0, 1)$ is not compact, and at this stage we know nothing about v: if $\beta v(kn)$ went to $+\infty$ as $n \to 1$ faster than $\log(1 - n)$ goes to $-\infty$, the maximum would not be attained. The same applies to the second form, where the choice set $[0, k)$ is not compact and $\log(1 - k'/k)$ goes to $-\infty$ as $k' \to k$.

2. Specify an appropriate state space, and show that the Bellman equation satisfies Blackwell's sufficient conditions for a contraction.

   Answer: Capital can only fall, since $k_{t+1} = k_t n_t \leq k_t$. Let $X = [0, k_0]$ (any $X = [0, \bar{k}]$ with $\bar{k} \geq k_0$ works as well). The state space is invariant: if $k \in X$ and $0 \leq n < 1$, then $kn \in [0, k] \subseteq X$. Let $B(X)$ be the space of bounded functions $f : X \to \mathbb{R}$ with the sup-norm $\|f\| = \sup_{k \in X} |f(k)|$, which is a complete metric space with $d(f, g) = \|f - g\|$. The Bellman equation defines the operator

   $$Tf(k) = \sup_{0 \leq n < 1} \{Ak + \log(1 - n) + \beta f(kn)\}.$$

We write sup instead of max because the choice set $[0,1)$ is not compact and f need not be continuous, so the maximum need not be attained.

T maps $B(X)$ into $B(X)$. Since $\log(1-n) \leq 0$, $Tf(k) \leq Ak_0 + \beta\|f\|$. Choosing $n = 0$ gives $Tf(k) \geq Ak + \beta f(0) \geq -\beta\|f\|$. Hence Tf is bounded. Note that the return function is unbounded below, since $\log(1-n) \to -\infty$ as $n \to 1$, but this does not matter: the planner can always choose $n = 0$.

The state space must be bounded. On $X = [0,\infty)$, T does not map bounded functions into bounded functions: for any $f \in B(X)$, choosing $n = 1/2$ gives

$$Tf(k) \geq Ak - \log 2 - \beta\|f\| \to \infty \quad \text{as } k \to \infty,$$

so Tf is unbounded. The same argument rules out $X = \mathbb{R}$, which in addition contains negative levels of capital.

We now verify that T satisfies Blackwell's sufficient conditions.

Monotonicity: Let $f, g \in B(X)$ be such that $f(x) \geq g(x)$ for all $x \in X$. Fix $k \in X$ and any $n \in [0,1)$. Since $f \geq g$,

$$Ak + \log(1-n) + \beta f(kn) \geq Ak + \log(1-n) + \beta g(kn).$$

The left-hand side is at most its supremum over n, which is $Tf(k)$, so $Tf(k) \geq Ak + \log(1-n) + \beta g(kn)$ for every $n \in [0,1)$. Taking now the supremum over n on the right-hand side gives $Tf(k) \geq Tg(k)$. Since $k \in X$ is arbitrary, $Tf(k) \geq Tg(k)$ for all $k \in X$. Hence, monotonicity is satisfied.

Discounting: For any $f \in B(X)$ and $a \geq 0$,

$$\begin{aligned} T(f+a)(k) &= \sup_{0 \leq n < 1} \{Ak + \log(1-n) + \beta\,(f(kn) + a)\} \\ &= \sup_{0 \leq n < 1} \{Ak + \log(1-n) + \beta f(kn)\} + \beta a \\ &= Tf(k) + \beta a. \end{aligned}$$

Hence, discounting is also satisfied. Since the Bellman operator satisfies Blackwell's sufficient conditions, it is a contraction mapping, with modulus β. By the contraction mapping theorem, it has a unique fixed point $v \in B(X)$.

3. Prove that the value function is increasing and convex, and explain the economic intuition underlying your result.

Increasing: It is sufficient to show that the Bellman operator preserves the property of a function being nondecreasing: if $f \in B(X)$ is nondecreasing, then Tf is nondecreasing too.

Why is this sufficient? By the contraction mapping theorem, the iterations $T^j f$ converge to the unique fixed point v from any initial guess $f \in B(X)$, uniformly

and hence pointwise. Start from a nondecreasing guess, such as $f = 0$. Since the Bellman operator preserves this property, each iteration is nondecreasing: for any $k_1 < k_2$ in X, $T^j f(k_1) \leq T^j f(k_2)$ for all j. Weak inequalities are preserved in the limit, so letting $j \to \infty$ gives $v(k_1) \leq v(k_2)$. Hence, the fixed point is nondecreasing.

To show that the Bellman operator preserves this property, let $f \in B(X)$ be nondecreasing, and let $k_1 < k_2$ be two levels of capital in X. For any $n \in [0, 1)$, $k_2 n \geq k_1 n$, so that $f(k_2 n) \geq f(k_1 n)$, and therefore

$$\begin{aligned} Tf(k_2) &\geq Ak_2 + \log(1-n) + \beta f(k_2 n) \\ &\geq Ak_1 + \log(1-n) + \beta f(k_1 n) + A(k_2 - k_1). \end{aligned}$$

Taking the supremum over n on the right-hand side gives $Tf(k_2) \geq Tf(k_1) + A(k_2 - k_1) \geq Tf(k_1)$.

Intuitively, if we start out with a higher initial capital level, we can keep our labor supply choices the same and reach strictly higher utility. Hence, optimally, we must also increase our utility if the initial capital level increases.

Convex: As for monotonicity above, it is sufficient to show that the Bellman operator preserves the property of a function being convex. That is, it is sufficient to prove that if $f(k)$ is convex, then $Tf(k)$ is convex too.

Why is this sufficient? Suppose that the Bellman operator preserves convexity. We have proven in subquestion 2 that the Bellman operator is a contraction mapping. Hence, by the contraction mapping theorem, the Bellman operator has a unique fixed point. Also, again by the contraction mapping theorem, the unique fixed point will be reached regardless of the initial "guess" of the value function. Hence, we can start out with an arbitrary convex function, such as $f = 0$. Since the Bellman operator preserves convexity, each iteration of the value function is convex. These iterations converge uniformly, and hence pointwise, to the fixed point. For any $k_1, k_3 \in X$ and $\lambda \in [0, 1]$, each iteration satisfies the weak inequality that defines convexity at these three points, and weak inequalities are preserved in the limit. Hence, the fixed point must also be convex.

To show that the Bellman operator preserves convexity, let $k_1 \in X$ and $k_3 \in X$ be two arbitrary levels of capital and let $k_2 = \lambda k_1 + (1 - \lambda) k_3$ for some $\lambda \in [0, 1]$. Also, let $f \in B(X)$ be an arbitrary convex function. Fix any $n \in [0, 1)$. Writing

k_2 explicitly and rearranging the terms that are linear in capital, we obtain

$$\begin{aligned}
&Ak_2 + \log(1-n) + \beta f(k_2 n) \\
&\quad = A\left(\lambda k_1 + (1-\lambda)k_3\right) + \log(1-n) \\
&\qquad + \beta f\left(\lambda k_1 n + (1-\lambda)k_3 n\right) \\
&\quad = \lambda\left(Ak_1 + \log(1-n)\right) + (1-\lambda)\left(Ak_3 + \log(1-n)\right) \\
&\qquad + \beta f\left(\lambda k_1 n + (1-\lambda)k_3 n\right) \\
&\quad \leq \lambda\left(Ak_1 + \log(1-n)\right) + (1-\lambda)\left(Ak_3 + \log(1-n)\right) \\
&\qquad + \beta\left(\lambda f(k_1 n) + (1-\lambda) f(k_3 n)\right) \\
&\quad = \lambda\left(Ak_1 + \log(1-n) + \beta f(k_1 n)\right) \\
&\qquad + (1-\lambda)\left(Ak_3 + \log(1-n) + \beta f(k_3 n)\right) \\
&\quad \leq \lambda T f(k_1) + (1-\lambda) T f(k_3),
\end{aligned}$$

where the first inequality follows from the convexity of f, and the last inequality follows from the fact that n does not necessarily maximize the right-hand side of the Bellman equation at k_1 and k_3. Since this holds for every $n \in [0, 1)$, taking the supremum over n on the left-hand side gives $Tf(k_2) \leq \lambda T f(k_1) + (1 - \lambda) T f(k_3)$. This shows that the Bellman operator preserves convexity. Therefore, the unique fixed point must be convex.

Intuitively, this result means that the higher the initial level of capital is, the larger the utility gains of a marginal unit of capital are.

4. Characterize the optimal decision rule as sharply as possible. Show that the capital stock reaches zero in finite time.

   Answer: Let $k^* \equiv 1/(A\beta)$. We show that the planner works if and only if $k > k^*$. Note that the objective $Ak + \log(1-n) + \beta v(kn)$ need not be concave in n, since v is convex, so we cannot simply check the first-order condition at $n = 0$.

   Below k^*, labor supply is zero. The interval $X^* = [0, k^*]$ is invariant, since $kn \leq k$, so we can solve the Bellman equation on X^* alone. Guess that $v(k) = Ak$ on X^*. With this guess, the objective is $Ak + \log(1-n) + \beta Akn$, which is strictly concave in n, with derivative

$$-\frac{1}{1-n} + \beta Ak \leq -1 + \beta Ak \leq 0 \qquad \text{for } k \leq k^*.$$

   Hence $n = 0$ is optimal, and the maximized value is Ak, which verifies the guess. By the uniqueness of the fixed point (on $B(X^*)$), $v(k) = Ak$ on X^*. Intuitively, the first unit of labor costs 1 in utility, the marginal disutility of labor at $n = 0$, and produces k units of capital, worth βAk in utility next period. For

$k \leq k^*$, this does not pay, and the capital stock is zero from the next period onward.

Above k^*, labor supply is strictly positive. Let $k > k^*$. For $n \leq k^*/k$, tomorrow's capital kn is at most k^*, so by the previous step $v(kn) = Akn$, and the objective is $Ak + \log(1-n) + \beta Akn$, whose derivative at $n = 0$ is $-1 + \beta Ak > 0$.

So the planner gains by moving away from $n = 0$: $n = 0$ is not optimal, and labor supply is strictly positive. Where the optimal n ends up does not matter for this argument: it may well be above k^*/k.

The capital stock reaches zero in finite time. The optimal n must do at least as well as $n = 0$, so that $\log(1-n) + \beta v(kn) \geq \beta v(0) = 0$. Since consumption is at most Ak_0 in every period and $\log(1-n_t) \leq 0$, $v(kn) \leq Ak_0/(1-\beta)$. Hence

$$\log(1-n) \geq -\frac{\beta A k_0}{1-\beta}, \qquad \text{that is,} \qquad n \leq \bar{n} \equiv 1 - e^{-\beta A k_0/(1-\beta)} < 1.$$

It follows that $k_t \leq \bar{n}^t k_0$, so the capital stock falls below k^* in finite time, and it is zero from the following period onward. Note that $n_t < 1$ alone would not be enough: a decreasing sequence of capital could converge to a strictly positive limit.

How labor supply varies with k above k^*. Assuming differentiability, the first-order condition $1/(1-n) = \beta k v'(kn)$ and the envelope condition $v'(k) = A + \beta n v'(kn)$ give

$$v'(k) = A + \frac{n}{1-n}\frac{1}{k}, \qquad \text{that is,} \qquad \frac{n}{1-n} = k\left(v'(k) - A\right).$$

For $k > k^*$, $n > 0$ and hence $v'(k) > A$. Since v is convex, $v'(k) - A$ is positive and nondecreasing, so $k\,(v'(k) - A)$ is strictly increasing in k and goes to infinity as $k \to \infty$. Hence, n is strictly increasing in k and $n \to 1$ as $k \to \infty$.

Additional remarks

Numerical solution. Using value function iteration, we can estimate the value function, the policy function, and labor supply numerically. Figure 3 confirms our theoretical findings.

A question to think about. The arguments in subquestion 3 show that v is nondecreasing and convex. Can they also show that v is strictly increasing? And strictly convex?

Answer: Strict monotonicity, yes: since $v = Tv$ and v is nondecreasing, the inequality $Tf(k_2) \geq Tf(k_1) + A(k_2 - k_1)$ with $f = v$ gives $v(k_2) - v(k_1) \geq A(k_2 - k_1) > 0$ for $k_2 > k_1$. Strict convexity, no: as shown in subquestion 4, $v(k) = Ak$ for $k \leq 1/(A\beta)$, which is linear.

Another question to think about. Utility is linear in consumption, and consumption is linear in capital. Why, then, is the value function convex rather than linear?

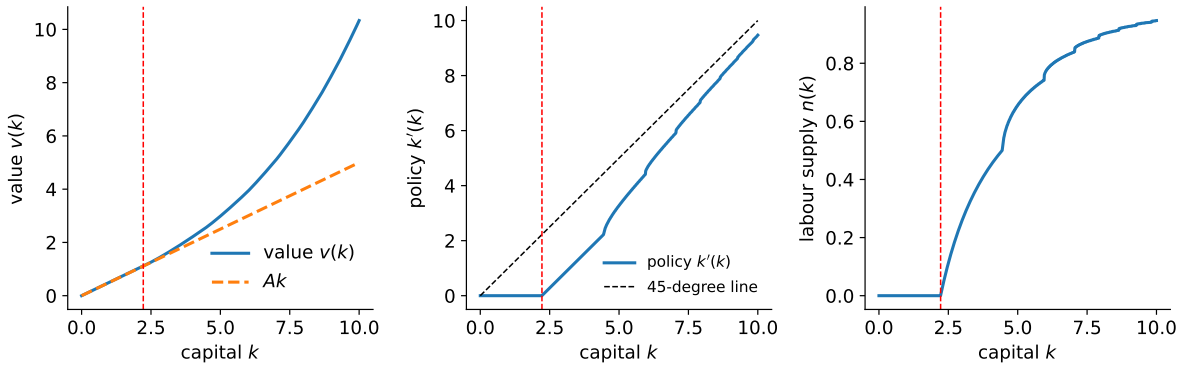


Figure 3: Value function, policy function, and labor supply in Question 3, for $A = 0.5$ and $\beta = 0.9$. The dashed red line is the cut-off $k^* = 1/(A\beta)$.